

Urbanization correlates with genetic and plastic variation of *Impatiens capensis* flower morphology

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Abstract

The spectacular diversity of flowers is largely driven by pollinator-mediated selection that favors attractive flowers and effective pollen transfer. Urbanization has the potential to affect floral trait evolution by altering pollinator communities through environmental changes. Additionally, abiotic changes in urban habitats can induce phenotypic plasticity, further shaping evolutionary trajectories. We investigated how urbanization affects the genetic and plastic components of flower morphology of *Impatiens capensis* (Balsaminaceae) across four Canadian cities. We found that urbanization influenced the pollinator community composition and the body size of bumblebees, the species' main pollinator, although the magnitude of the size effect varied among cities. Using a combination of field surveys and a common garden experiment, our results suggest that urbanization affects sepal size—a tubular floral organ in which pollinators enter to access the nectar—through both genetic and plastic responses. While plasticity sometimes masked the genetic determination of sepal size in the field, we observed a positive correlation between the genetic component of the sepal size and bumblebee body size. These results suggest that urban habitats may drive evolutionary changes in floral traits by modifying pollinator communities.

Keywords plant-pollinator interactions, pollinators, flower size, flower shape, phenotypic plasticity, bumblebee size

Introduction

Flowers show an astounding diversity in shapes, sizes, colors, and scents across angiosperms, mostly resulting from selection imposed by pollinators (Galen, 1999; Harder & Johnson, 2009; Ollerton et al., 2011; Stebbins, 1970; van der Niet & Johnson, 2012; Whittall & Hodges, 2007). Flowers can genetically adapt to changes in pollinator communities or abundance by evolving traits that enhance attraction, such as color, scent, and nectar, that facilitate pollen transfer by improving the mechanical fit with pollinators through shape or size modifications, or that promote self-pollination (Acoca-Pidolle et al., 2024; Gervasi & Schiestl, 2017; Gómez et al., 2009). But flowers can also be modified by their environment through phenotypic plasticity (Caruso, 2006; Gómez et al., 2020), which can be either adaptive or maladaptive depending on whether the environment modifies the phenotype in a way that increases or decreases the mean fitness of the population, respectively (Dorey & Schiestl, 2022). Also, phenotypic plasticity can hide genetic adaptation if it af-

fects the phenotype in a way that opposes natural selection, a concept called counter-gradient variation (Conover & Schultz, 1995; Conover et al., 2009). Despite the important roles of genetics and phenotypic plasticity in determining phenotypes, few studies have tried to tease them apart when investigating evolutionary responses of flower morphology across environmental gradients.

Urban habitats offer a unique opportunity to study evolutionary responses to environmental changes (Johnson & Munshi-South, 2017). The development and spread of cities alter abiotic conditions by remodeling the physical environment and biotic conditions through their effects on species richness, abundance, interactions, and evolution (Alberti et al., 2017; Johnson & Munshi-South, 2017; McKinney, 2008; Santangelo et al., 2022). Furthermore, since urban habitats are generally more similar to each other than to their surrounding natural environments (Santangelo et al., 2022), studying species response to urbanization in different cities is equivalent to a replicated evolutionary experiment (Santangelo et al., 2020), and could uncover parallel or idiosyn-

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cratic responses across cities (Diamond et al., 2018; Santangelo et al., 2022).

Urbanization is expected to disrupt mutualistic interactions, such as pollination, because these interactions require compatibility between flowering and pollinator activity and phenotype matching between the plant and pollinator traits (Fisogni et al., 2020; Irwin et al., 2020). Pollinator communities commonly exhibit changes along urbanization gradients (Baldock et al., 2019; Hall et al., 2017; Theodorou et al., 2020; Wenzel et al., 2020), but urban habitats could also affect the body size of insect pollinators via changes in the pollinator species composition or within species through evolutionary responses to changes in resource availability and temperature (Goulson, 2010). So far, studies have produced conflicting results regarding how urbanization affects bumblebee size (Eggenberger et al., 2019; Theodorou et al., 2021). A greater distance between resource patches, as commonly found in cities, could favor larger insects with greater flight capabilities (Greenleaf et al., 2007). On the other hand, the high temperatures of cities could favor smaller insects with greater thermoregulation potential (Eggenberger et al., 2019). Such changes in pollinator body size, as well as those in the abiotic environments, can potentially affect flower morphology through both genetic and plastic responses. Yet, few studies have investigated the impact of urbanization on the evolution of flower traits while also considering changes in pollinator communities (Irwin et al., 2018; Rivkin et al., 2020; Ushimaru et al., 2014) and attempted to disentangle genetic from plastic responses (but see Irwin et al., 2014).

Impatiens capensis Meerb. (Balsaminaceae) is an excellent system for studying the evolution of flowers due to its reported capacity to respond to its pollinator communities in both urban and natural habitats. This annual species has a mixed pollination strategy, bearing both closed, green, and tiny self-fertilizing (cleistogamous) flowers, and, open, orange, large protandrous and mainly outcrossing (chasmogamous) flowers (Figure 1). This reproductive strategy is plastic, as a greater proportion of chasmogamous flowers are produced with more light (Schemske, 1978; Waller, 1980), but it does not seem to be affected by urbanization (Barker & Sargent, 2020). Populations of *I. capensis* growing in sunlight produce numerous chasmogamous flowers pollinated by bumblebees, other bees, wasps, flies, and hummingbirds (Fenster et al., 2004; Rust, 1977, 1979), which all feed on pollen or the nectar secreted in the spur. There has been extensive research on *I. capensis* over the past 50 years, and a fair amount of investigation into its flower morphology, although few studies span its whole distribution (Beans & Roach, 2015; McDonald & Caruso, 2019; Panique & Caruso, 2020; Travers et al., 2003; Young, 2008; Zhao & Schoen, 2022). The spur curvature of *I. capensis* is heritable, and affects pollen removal by hummingbirds (Travers et al., 2003) and the duration of visits by bees (Young, 2008). *Impatiens capensis* evolved smaller sepals under relaxed pollinator-mediated selection (Zhao & Schoen, 2022), and larger flowers under pollinator limitation (Panique & Caruso, 2020). Also, natural populations in Montreal, Canada, had larger sepals and longer nectar spurs compared to their urban counterparts (Faure et al., 2023). Finally, reproductive success was shown to be limited by pollen deposition in urban and natural habitats (Barker & Sargent, 2020), potentially allowing natural selection to drive floral phenotypic changes (Harder & Aizen, 2010).

We investigated whether urbanization impacts, directly or indirectly through its effects on pollinator communities, the size

and shape of flowers of the spotted jewelweed, *I. capensis*. Specifically, we tested four hypotheses. First, we tested whether the pollinator community of *I. capensis* differed between urban and natural habitats given that previous studies have found differences in community composition and pollinator sizes. Second, we tested whether sepal size is positively correlated with pollinator size given that pollinators of *I. capensis* must pass through the large tubular posterior sepal to access the nectar produced in the spur, and because the reproductive organs are situated at the top entrance of the sepal. Third, we tested whether corolla surface is negatively correlated with the visitation rates of pollinators given that previous studies have shown that increased competition for floral resources (Hennig & Ghazoul, 2011) and reduced pollen dispersal in urban habitats (Rivkin et al., 2020) should result in stronger pollen limitation (Zink et al., 2024). Consequently, selection should favor larger floral displays in pollinator-limited populations to attract more pollinators (Geslin et al., 2013; Thomann et al., 2013). Fourth, we tested whether sepal and corolla shapes differ between urban and natural habitats. We expect sepal shape to differ to maximize pollen transfer, but not corolla shape, as size appears more important than shape for pollinator attraction.

To test these hypotheses, we performed both field investigations and a greenhouse common garden experiment to control for environmental effects. We studied populations of *I. capensis* growing in sunlight to focus on the evolutionary responses of their animal-pollinated flowers. We considered the phenotypes measured in the common garden as the genetic components of the corresponding traits, whereas the phenotypes measured in the field corresponded to joint genetic and plastic components of the traits. We quantified flower shapes and sizes using geometric morphometrics, and we characterized pollinator communities for a total of 16 populations, paired by habitat—urban versus natural—in and around four large Canadian cities (“regions” hereafter): Toronto, Ottawa, Montreal, and Quebec (Figure 1A).

Materials and methods

Population selection and characterization

To test whether urbanization impacted the size and shape of *I. capensis* flowers, we compared urban and natural populations of four large Canadian metropolitan areas (> 500,000 inhabitants). We selected two pairs of urban and natural populations with more than 100 individuals growing in open canopies in an area of 75 km around the city center of each region, for a total of 16 populations (Table S1). Urban populations were selected to have high proportions of impervious surfaces, while natural populations were selected to have fewer impervious surfaces, more vegetation and few crop fields around them. One natural population selected in 2021 in the Toronto region produced too few chasmogamous flowers and was abandoned (only 15 populations were sampled in 2021). It was replaced with the Silver Creek population in 2022 (Table S1). The Oka natural population in the Montreal region in 2021 was replaced by the Mirabel population in 2022 because of access issues (Table S1).

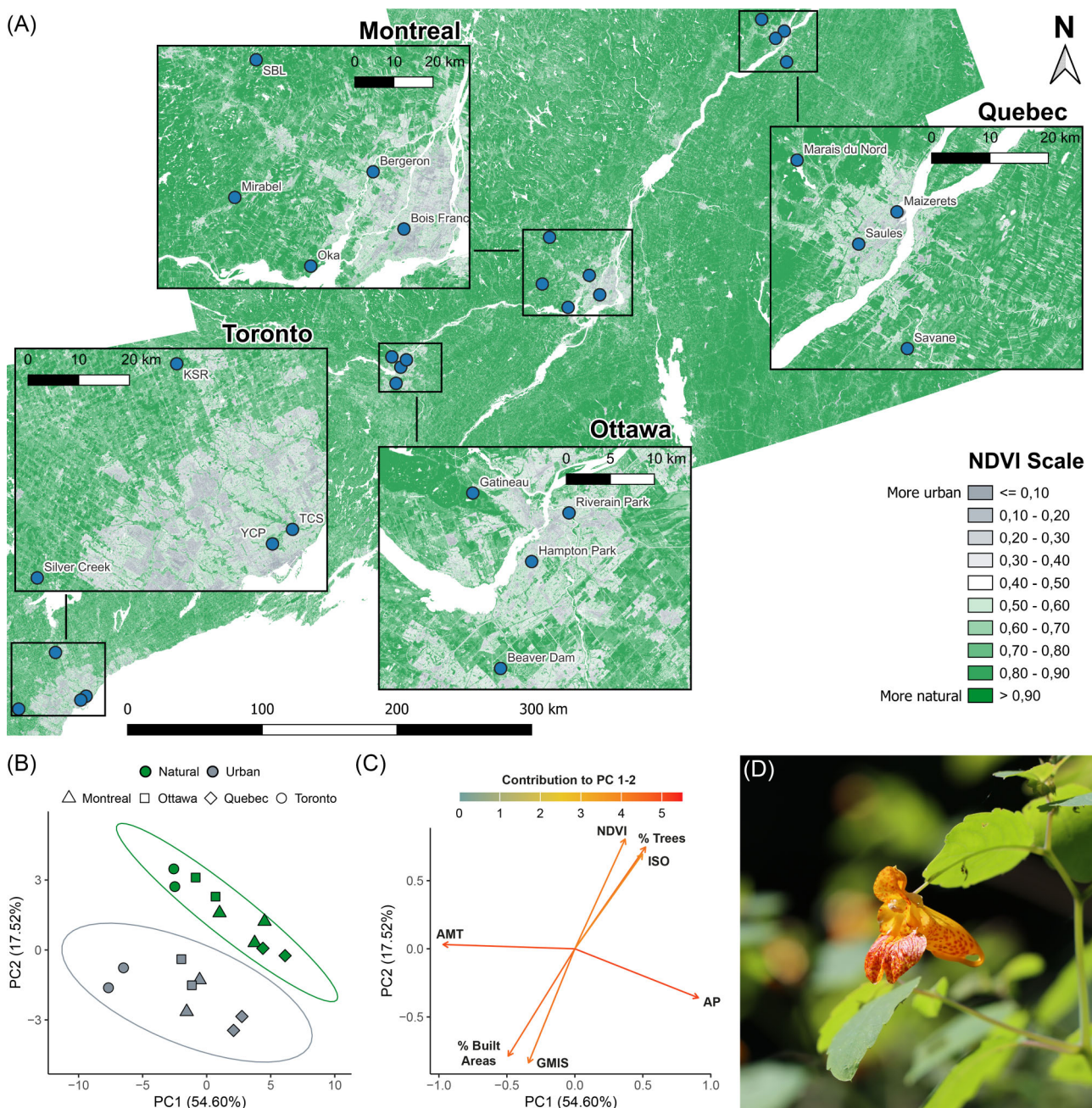


Figure 1. Study sites and system. Flowers from populations of *Impatiens capensis* were sampled in four regions in 2021 and 2022. (A) Normalized Difference Vegetation Index (NDVI) map showing the sampled populations; high NDVI values indicate more important vegetation cover. More information about populations can be found in [Table S1](#). (B) Populations and (C) variables plots of a Principal Component Analysis (PCA) conducted on the mean values of environmental variables from a 500m radius circle around each population. Only the most contributing variables are represented: annual mean temperature (AMT), annual precipitation (AP), isothermality (ISO), NDVI, global man-made impervious surfaces (GMIS), percentage of trees (% Trees) and percentage of built areas (% Built Areas). Ellipses represent 95% confidence interval (CI). PCAs for 1000m and 2000m radius buffers are available in the [Supplementary Material](#). (D) Flower of *Impatiens capensis* (credits: J. Burkiewicz). Flowers are typically 15–30 mm long and 15–40 mm high.

To confirm that urban and natural sites were clearly distinct across regions, we characterized each population with the Normalized Difference Vegetation Index (NDVI) and the percentage of land use/land cover using satellite images from Sentinel-2 (resolution 10m), and the mean Global Man-made Impervious Surface (GMIS) ([Brown de Colstoun et al., 2017](#)) using Landsat 7 data (resolution 30m). We measured the mean NDVI, GMIS, and land use percentages for each population with buffer zones of 500

m, 1000 m, and 2000 m radius centered on them. We also characterized each population using bioclimatic variables. We interpolated values of temperature (minimum and maximum, mean, mean minimum and maximum, mean dew), precipitation (total, mean relative humidity, number of wet and dry days), solar radiation, and frost-related metrics (number of days with and without) using BIOSIM 11 ([Régnière & Bolstad, 1994](#)), and computed the monthly summary for each population between

2010 and 2022 to obtain average bioclimatic conditions across several years for each population. These values were used to compute worldclim bioclimatic variables using the QBMS package (Al-Shammas, 2021) for all populations in R version 4.3.3 (R Core Team, 2024). To compare the populations, we ran a scaled principal component analysis (PCA) including all above abiotic variables.

Field sampling

In both 2021 and 2022, we visited each population to photograph flowers, collect pollinators, and record flower visitors with cameras, and a second visit allowed us to collect pollinators and record flower visitors again. Both visits were done during the flowering season (between August 1st and September 10th). In each population, we photographed 50 flowers in profile and front view from randomly selected plants with either a Canon T2i with a EF-S 60mm macro lens (1:2.8 USM, Canon Inc., Tokyo, Japan) or a Canon 90D with a EF 100mm macro lens (1:2.8 L IS USM, Canon Inc., Tokyo, Japan), placing a ruler beside flowers. We also recorded flower visitors of at least five different flowers for one hour using two GoPro Hero 9 cameras (GoPro, San Mateo, California, USA) placed 2 to 50 m apart. The cameras were positioned 30 to 50 cm from the flowers, and recordings were done in 4K (3820 × 2160 pixels), with 30 frames per second. In 2021, we also sometimes recorded pollinator visits using a Canon T2i with a EF-S 60mm macro lens in 1080p (1920 × 1080 pixels) at 30 frames per second.

After the cameras stopped recording, two people collected insect pollinators of *I. capensis* for further identification and measurements during 20-minute sessions, using entomological nets while standing in front of, or walking along a stand of *I. capensis*. Pollinators were caught if they touched the reproductive parts of the flower and were then placed in a 50 mL tube in a cooler and frozen at the end of the day. All pollinator observations and captures were conducted in sunny conditions, with a minimum temperature of 15°C and low to no wind.

Common garden experiment

During the summer of 2021, we collected one seed from at least 30 plants in each population for a common garden experiment. A few days after collection, seeds were stored in Petri dishes on moist filter paper, closed with parafilm, and maintained at 4°C until germination (ca. 4–5 months later). Seedlings were transferred to 2-inch square pots containing BM6 multipurpose soil (Berger, Saint-Modeste, Quebec, Canada) and watered every two days from the bottom of the tray, alternating between clear water and fertilized water (15–15 NPK fertilizer from Monday to Thursday and 19–2–19 NPK from Friday to Sunday). When the plants reached 15 cm in height, they were transferred to 8.5-inch round pots with BM6 soil. They were watered and fertilized every two days, and every day from March 1st. Plants were grown with 16 hours of light, with day and night temperatures of 22°C and 18°C, respectively. When plants started flowering, we photographed the front and profile views of one random flower on each plant for phenotyping, using a Canon T2i with an EF-S 60mm macro lens, including a ruler on each picture (Table S1). The common garden experiment aimed to remove the environmental effects on the plants' phenotypes, but

because it consisted of the progeny of the natural plants, maternal effects might be expressed.

Pollinator identification and measurement

Frozen pollinators were pinned on foam blocks using needles and dried at room temperature for a week. Apart from one Calliphoridae, one Muscidae, and one wasp specimen (the last being identified at the genus level), pollinators were identified at the species level using the following identification keys: Ascher & Pickering, 2011; Bohart et al., 1976; Buck et al., 2008; Sheffield et al., 2011; Skevington et al., 2019; Williams et al., 2024. Pollinators were deposited at the Ouellet-Robert entomological collection of the Université de Montréal (Burkiewicz, 2025). We used the inter-tegular distance (ITD) as a proxy of pollinator size, as it is a good indicator of insects' dry mass and flight capability (Goulson, 2010; Greenleaf et al., 2007), and because it is correlated with proboscis length in bees (Cariveau et al., 2016), an important trait to reach the nectar concealed in the spur. We measured it by photographing each pollinator under a Zeiss microscope equipped with an AxioCam HRC (Discovery V20) camera and using the software ZENblue v2.5 Pro (Carl Zeiss, Oberkochen, Germany).

Pollinator observations

On the camera recordings, we identified between 5 and 10 open flowers from which we could easily distinguish a pollination event. An animal was considered a pollinator when it touched the reproductive parts of the flowers. Videos were watched at 2 × speed. When an animal was observed, videos were slowed down (0.12–0.5 ×) to score pollination events. For each pollinator, we recorded the time of the bout, the species or genus and family, the number of flowers observed in this particular recording, and the number of flowers visited. We identified bumblebees at the genus level, other bees at the species level, and wasps and flies at the family level. When we could not determine the species or genus, we identified taxa at the family level.

We calculated the total observation effort as the number of hours observed times the number of flowers observed for each camera recording. The pollinator visitation rate was the number of flowers pollinated over the total observation effort. Finally, the relative frequency of each pollinator taxon was the number of pollination events observed for that taxon divided by the total number of pollination events, either at the population level or over all populations.

Flower shape and size characterization

We used geometric morphometrics to characterize the flower shape from flower pictures taken in profile and front views, and measure the posterior sepal size and corolla surface (Collyer et al., 2015; Klingenberg, 2010; Mitteroecker & Gunz, 2009). We used tps-Dig v2.32 (Rohlf, 2021) to digitize specimens. We used a similar approach to that of Faure et al. (2023) to extract flower shape, and details on landmarks and semilandmarks positioning are available in Supplementary Materials. Flower size was estimated from the landmarks and semi-landmarks that delimited the contour of the sepal and the corolla; more details can be found in the Supplementary Materials. To compare flower shape between populations we performed a General Procrustes Analysis (GPA) with the *gpa*gen

function of the geomorph package (Adams et al., 2020), minimizing Procrustes distance between individuals (Klingenberg, 2010). As in Faure et al. (2023), semilandmarks were treated as landmarks in the GPA because the high variability of the spur and petals' curvature caused misalignments when treated as semilandmarks. We manually checked for outlier shapes to ensure these were not due to improper landmark positioning.

To assess the technical error in landmark positioning, we placed landmarks on duplicate flowers in some populations (four in front view, five in profile view). The technical error estimated using a Procrustes ANOVA with 999 permutations (*procDlm* function of the geomorph package (Adams et al., 2020) was 0.02% for the profile pictures and 2.08% for the front pictures.

Statistical analyses

All analyses were performed in R version 4.3.3 (R Core Team, 2024) using the *car* (Fox et al., 2001), *emmeans* (Lenth, 2017), *lme4* (Bates et al., 2003), *lmerTest* (Kuznetsova et al., 2017), *geomorph* (Adams et al., 2020), and *vegan* (Oksanen et al., 2025) packages. We checked model's assumptions and fit (normality of residuals, homoscedasticity) visually, using diagnostic plots (Q-Q plot for normality of residuals, residuals vs. predicted plot for homoscedasticity), as indicated in Zuur et al. (2009). In all analyses, we used sum-to-zero contrasts to compute the summaries of the models.

Pollinator abundance—We evaluated whether pollinator communities differed between urban and natural habitats. Because observation data and collected specimens gave consistent results (Figure 2A, Table S2), we used only collected specimens for statistical analysis, as most insects were identified at the species level. Redundancy analysis (RDA) was used to analyze community data rather than canonical correspondence analysis (CCA) because in the latter, rare species contribute importantly to the chi-square distance (Legendre & Legendre, 2012), and we were more interested in differences in the main pollinators. We used a Hellinger transformation of the data to account for shared absences among sites (Legendre & Legendre, 2012). We tested whether regions (cities) or habitats (urban/natural) could explain variation in pollinator abundance. Partial RDA was used to test each factor while removing the effect of the other.

Pollinator visitation rate—We assessed whether urbanization influenced pollinator visitation rates. We used region, habitat and their interaction as fixed factors in an ANOVA to explain the variation in pollinator visitation rate, assuming a Gaussian distribution. Significance was tested using the *Anova* function from the *car* package and type II sum of squares. We calculated the coefficients of the model and the associated P-values using the *summary* function.

Bumblebee size and flower size—We tested whether urbanization affected the sizes (ITD) of bumblebees (all species combined), the main pollinator of *I. capensis*, and of flowers (sepal size and corolla surface). We built a linear mixed model that included Year (when applicable), Region, Habitat, and their interactions as fixed effects, including a population random effect and allowing slopes to vary with region, habitat, and their interaction:

$$\text{Size} \sim \text{Year} \times \text{Habitat} \times \text{Region} + (\text{Region} \times \text{Habitat} | \text{Population})$$

We first selected the best variance structure by fitting and comparing models with random effects of descending order of com-

plexity and all fixed effects. Models were fitted with Restricted Maximum Likelihood (REML) using the *lmer* function from the *lme4* R package and the best variance structure was the one with the lowest AIC (Zuur et al., 2009). Singular models and models that did not converge were excluded from the comparisons. The best variance structure for the three response variables was a simple population random effect, independent of region or habitat:

$$\text{Size} \sim \text{Year} \times \text{Habitat} \times \text{Region} + (1 | \text{Population})$$

Using this variance structure, we then tested fixed effects and their interactions using type II Wald tests (*Anova* function from the *car* package). We calculated the coefficients of the model and the associated P-values (with Kenward-Roger degrees of freedom) using the *summary* function. We also computed estimated marginal means contrasts for urban/natural comparisons within regions within years using the Holm correction for multiple testing to present trends per region.

For flower size, we also tested if the sexual stage of the flowers (male or female) was correlated with the sepal size or the corolla surface, thus eliminating a potential confounding effect. We used an ANOVA with flower size as the response variable and sex (male or female) as the explanatory variable to test this correlation. The sexual stage was weakly associated ($R^2_{\text{adj}} < 1\%$) with sepal size and corolla surface in the field. To further test the robustness of the results to the sexual stage, we compared the results of models that included or excluded female flowers (12.2% of the data) and found consistent results for both analyses (Table S6), indicating that our results were not affected by the sexual stage of the flower.

Plasticity of flower size—We tested whether flower size was affected by phenotypic plasticity and if this effect was influenced by urbanization by comparing the flower size measured in the field to that measured in the common garden. We first analyzed the data from the 2021 field season versus the common garden experiment to compare the phenotype of the parental population and the phenotype of the offspring generation (the parentage of which is unknown), showing both genetic and plastic responses to environmental change. We then analyzed the data from the 2022 field season versus the common garden experiment to compare the phenotypes of kin, representing the plastic response to environmental change.

For each comparison, we built a linear mixed model to analyze variation in flower size with Environment (either field or common garden), Habitat (either urban or natural), and their interaction as fixed effects, including a population random effect, allowing slopes to vary with environment, habitat, and their interaction. We used the same procedure mentioned above to select the best variance structure. The best model included a population random effect that changed the intercept and the slopes were allowed to vary with Environment:

$$\text{Size} \sim \text{Environment} \times \text{Habitat} + (\text{Environment} | \text{Population})$$

We then tested fixed effects and interactions using type II Wald tests using the *Anova* function from the *car* package. We calculated the coefficients of the model and the associated P-values (with Kenward-Roger degrees of freedom) using the *summary* function.

Correlation between bumblebee size and sepal size, and between pollinator visitation rates and corolla surface—We tested

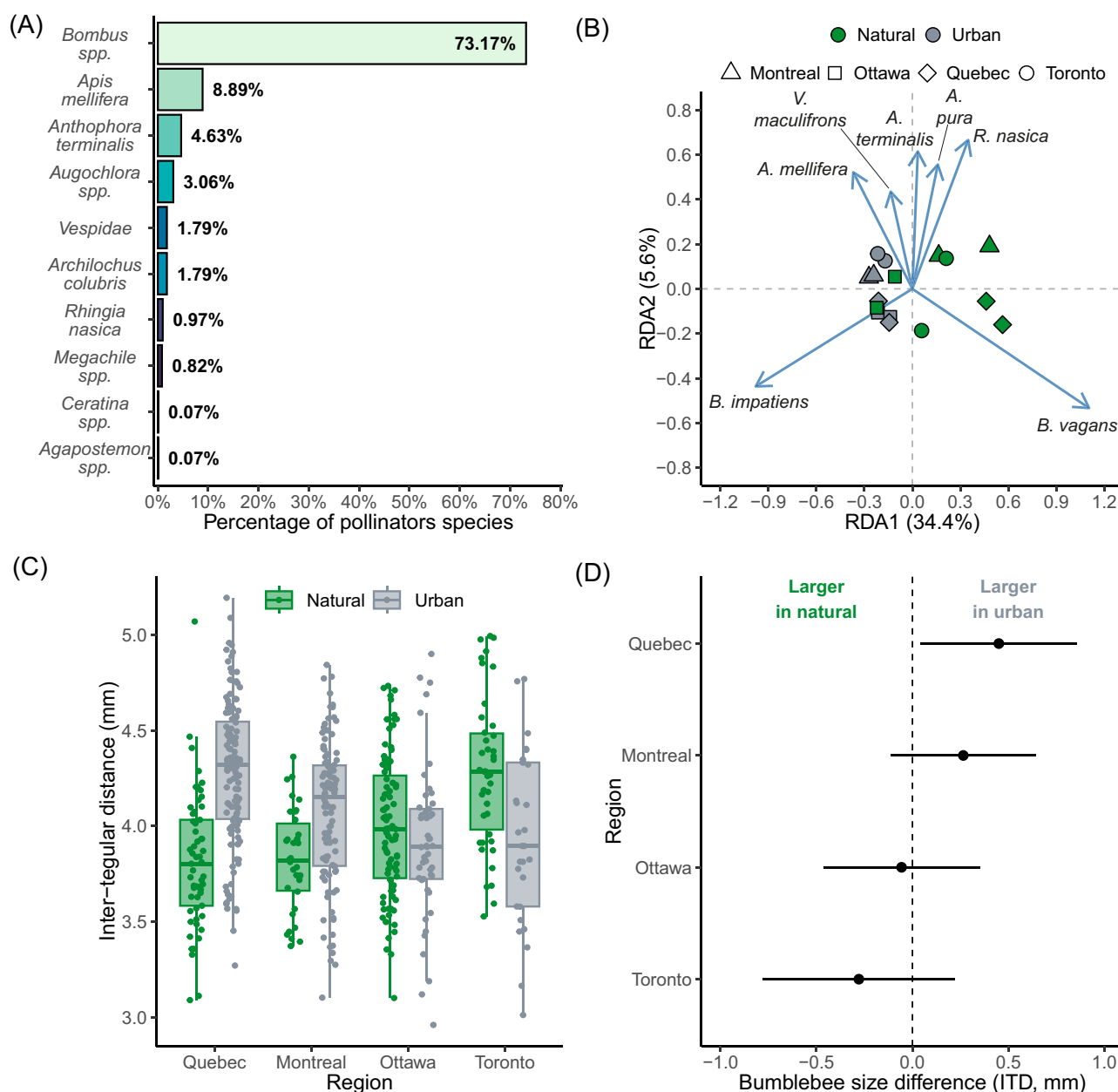


Figure 2. Characterization of the *I. capensis* pollinator community and size of bumblebees. (A) Relative frequencies of pollinator species computed from video observation data, although frequencies are similar to those obtained with collected specimens (Table S2). (B) Redundancy analysis of pollinator abundance (collected specimens) by habitat and region for the 2022 field season. Only the most abundant species were represented on the graph for clarity purposes. The redundancy analysis on combined field seasons (2021 and 2022) is available in Figure S2. (C) Boxplots of bumblebee inter-tegular distances by habitat and region, averaged over the two field seasons. Dots represent individual bumblebee measurements; sample sizes are available in Table S4. The boxplot of *B. impatiens* only is available in Figure S3. (D) Estimated marginal means contrasts of bumblebee size between urban and natural habitats for each region from the linear mixed model and associated 95% confidence intervals. Measures of bumblebee size include all *Bombus* spp. collected.

whether variation in mean population sepal sizes was correlated with the variation in mean population bumblebee sizes (ITD). We used the sepal sizes measured in the common garden because they correspond to the genetic component of the trait, and analyzed their variation with an analysis of covariance with habitat and ITD and their interaction as fixed factors:

$$\text{Sepal Size} \sim \text{Habitat} \times \text{ITD}$$

We tested whether variation in mean population corolla surfaces was correlated with the variation in mean population visitation rates. We used the corolla surfaces measured in the common garden because they correspond to the genetic component of the trait, and analyzed their variation with an analysis of covariance with habitat and the average pollinator visitation rates and their interaction as fixed factors:

$$\text{Corolla Surface} \sim \text{Habitat} \times \text{Pollinator Visitation Rates}$$

Significance was tested using the *Anova* function of the *car* package and type II sum of squares as above.

Flower shape—We used a Procrustes ANOVA (equivalent to a PERMANOVA) with the *procD.lm* function from the *geomorph* package (Adams et al., 2020) to test whether flower shape was influenced by size (allometry effect), year (when applicable), sexual stage of the flower (male vs female), region, and habitat, including all possible interactions between these variables:

$$\text{Shape} \sim \text{Size} \times \text{Year} \times \text{Sex} \times \text{Region} \times \text{Habitat}$$

Because Procrustes ANOVA cannot include random factors, we followed Collyer et al. (2015) and used the interaction of habitat and population as the error term to test for fixed effects on interpopulation variation. We removed all non-significant interactions, except the one between habitat and region, which we were interested in, and ran a Procrustes ANOVA with the remaining terms. We used type 1 Procrustes ANOVA, ordering factors in a way that we tested for the effect of habitat and the interaction between habitat and region after all other factors.

Results

Urban populations were environmentally distinct from natural populations

Urban and natural populations, across the four regions studied, were characterized by different abiotic environments, showing our ability to discriminate those environments in the field and justifying the use of the “habitat” categorical variable in all analyses that quantify the effects of urbanization (Figure 1 and Figure S1). The natural populations were associated with higher tree cover and greater vegetation index (NDVI), whereas urban populations were associated with higher proportions of impervious surfaces (GMIS) and built areas, as well as lower isothermality (day-to-night temperature variation compared to annual oscillations) due to a heat island effect (Figure 1B, 1C and Figure S1). The regions were differentiated along by their annual mean temperature and annual precipitation, following a southwest to northeast gradient with colder and wetter cities in the northeast (Figure 1B, 1C, and Figure S1).

Urbanization modifies the pollinator communities of *I. capensis*

Bumblebees (*Bombus* spp.) were the most frequent pollinators of *I. capensis*, representing 73% of all pollination events recorded on video (Figure 2A). *Bombus impatiens* was the most collected species (62.4%, Table S2), followed by *Bombus vagans* (12.7%, Table S2). *Apis mellifera* represented 6.4% of the specimens sampled (Table S2) but primarily collected pollen and may not be a reliable pollinator, as stigmas are not receptive when anthers are mature. *Anthophora terminalis*, *Vespidae*, and *Augochlora pura* together represented 12.5% of the specimens collected (Table S2), and these less common and less cosmopolitan pollinators did not make contact as consistently with the reproductive structures as bumblebees, according to video recordings.

Pollinator communities of *I. capensis* differed between urban and natural habitats ($R^2_{\text{adj}} = 0.264$, $P = 0.013$; Figure 2B, Figure S2). Urban habitats were more associated with *Bombus impatiens*, *Vespula maculifrons*, and *Apis mellifera*, while natural habitats were more associated with *Bombus vagans*, *Augochlora pura* and *Rhingia nasica* (Figure 2B, Figure S2). Among these pollinators, bumblebees were the most consistently found across all populations, and as such, they best characterized urban and natural pollinator communities of *I. capensis*. We also found that the effect of urbanization on pollinator visitation rates varied among regions (Region $\times$ Habitat: $F_{(3, 23)} = 3.675$, $P = 0.027$; Table S3A, S3B). Visitation rate tended to be higher in urban habitats of the Quebec and Montreal regions (76.1% and 37% increases, respectively, from natural habitats), whereas it tended to be higher in natural habitats of the Ottawa and Toronto regions (43.8% and 27.5% increases, respectively, from urban habitats), although none of these comparisons were statistically significant in post-hoc tests (Table S3C).

Finally, we found that the effect of urbanization on bumblebee size varied among regions (Region $\times$ Habitat: $\chi^2_{(3)} = 8.827$, $P = 0.031$; Figure 2C, 2D, Table S5A, S5B). Bumblebees exhibited larger sizes in urban habitats of the Quebec and Montreal regions compared to the natural ones (11.6% and 6.8% increase, respectively), whereas the trend was opposite in the Toronto region (6.7% decrease), although none of these post-hoc comparisons were statistically significant, and sizes were similar in the Ottawa region (Figure 2C, 2D, Table S5C). Results were similar when analyzing only *B. impatiens* (Figure S3), showing that differences in size among habitats of some regions were probably due to both intraspecific and interspecific differences as *B. impatiens*, more present in urban environments, appears slightly larger than *B. vagans* (*B. impatiens* size = $4.11 \text{ mm} \pm 0.019$, $N = 432$; *B. vagans* = $3.84 \text{ mm} \pm 0.031$, $N = 88$).

Urbanization affects the genetic component of sepal size

There were no significant differences in sepal size between urban and natural populations in the field ($\chi^2_{(1)} = 0.240$, $P = 0.624$; Figure S4, Table S6). However, in the common garden experiment, sepals from urban populations were on average 7.6% larger than those in natural populations ($\chi^2_{(1)} = 8.564$, $P = 0.003$; Table S7A), though this effect varied among regions (Region $\times$ Habitat: $\chi^2_{(3)} = 8.329$, $P = 0.040$; Table S7A, S7B). Urban sepals were larger in the region of Quebec and Montreal compared to the natural ones (20.3% and 7.9% increases, respectively), although only the comparison in Quebec was statistically significant in post-hoc tests, and no differences were found in the Ottawa and Toronto regions (Figure 3B, 3C, Table S7C). Together, these results suggest that phenotypic plasticity in the field could hide the genetic differences in sepal size between habitats. To further investigate this phenotypic plasticity, we analyzed the mean response of sepal size per habitat to changes in the environment (field vs. common garden). Compared to the common garden experiment, the sepal size measured in the field was smaller in urban populations (2.4% reduction for the 2021 comparison, and 4.1% reduction for the 2022 comparison) and larger in natural populations (7.1% increase for the 2021 comparison, and 4.4% increase for the 2022 comparison), thus

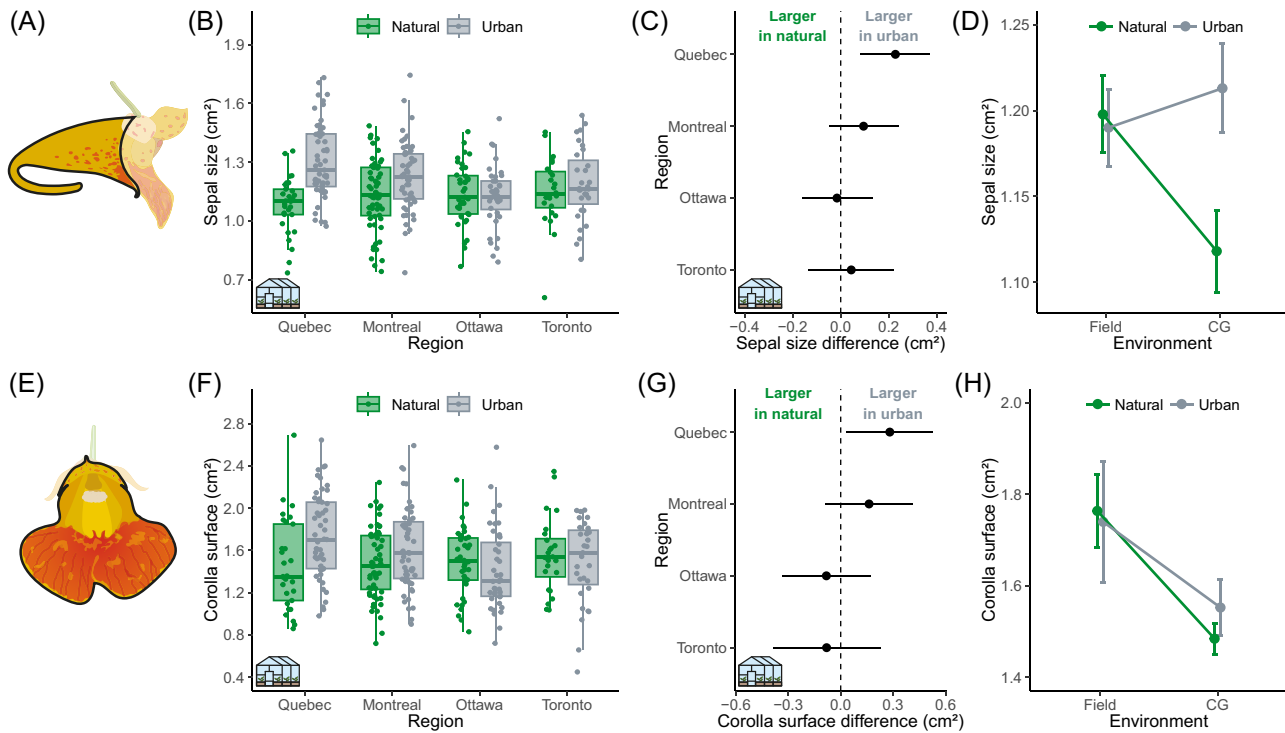


Figure 3. Sepal size (A–D) and corolla surface (E–H) comparisons between habitats. (A) Drawing showing the measure used to estimate the sepal size of *Impatiens capensis* (black outline). (B) Boxplots of the sepal size measured in the common garden (as indicated by the greenhouse icon) with dots representing individual flowers. (C) Estimated marginal means contrasts of the sepal size between urban and natural habitats per region and associated 95% confidence intervals in the common garden experiment. (D) Mean response of sepal size, grouped by habitats, to environmental change for the 2021 field season. Error bars represent standard errors of the mean. Results with two field seasons combined are available in [Figure S5A](#). (E) Drawing showing the measure used to estimate the corolla surface of *Impatiens capensis* (black outline). (F) Boxplots of the corolla surface measured in the common garden with dots representing individual flowers. (G) Estimated marginal means contrasts of the corolla surface between urban and natural habitats per region and associated 95% confidence intervals in the common garden experiment. (H) Mean response of corolla surface, grouped by habitats, to environmental change for the 2021 field season. Error bars represent standard errors of the mean. Results with two field seasons combined are available in [Figure S5B](#). Credits for the drawings: J. Lafond.

completely masking the existing genetic differences in sepal size between habitats (Environment $\times$ Habitat, 2021: $\chi^2_{(1)} = 4.796$, $P = 0.028$; 2022: $\chi^2_{(1)} = 3.707$, $P = 0.054$; [Figure 3D](#), [Figure S5A](#), [Table S8](#)).

We found that the genetic component of the mean sepal size per population was positively correlated with the mean population bumblebee size ($F_{(1,11)} = 6.531$, $P = 0.026$, $R^2_{\text{adj}} = 0.484$; [Figure 4](#), [Table S9](#)).

Urbanization has a limited effect on corolla surface, which is strongly affected by phenotypic plasticity

In the field, corolla surface was affected by habitat, but only in certain regions and the effect of urbanization also varied between years, suggesting phenotypic plasticity ([Figure S6](#), [Table S6](#)). Under controlled conditions, the effect of urbanization on corolla surface varied among regions (Region $\times$ Habitat: $\chi^2_{(3)} = 8.136$, $P = 0.043$; [Figure 3F](#), [3G](#), [Table S7A](#), [S7B](#)). Urban corollas were larger in the region of Quebec and Montreal (19.1% and 10.8% increases, respectively), and smaller in Ottawa and Toronto regions (5.3% and 5.1% decrease, respectively) compared to the natural ones, although none of these comparisons were statistically sig-

nificant in post-hoc tests ([Figure 3F](#), [3G](#), [Table S7C](#)). The corolla surface was 19% larger in the field compared to the common garden, regardless of habitat, confirming strong phenotypic plasticity for this trait (2021: $\chi^2_{(1)} = 8.860$, $P = 0.003$; 2022: $\chi^2_{(1)} = 36.761$, $P < 0.001$; [Figure 3H](#), [Figure S5B](#), [Table S8](#)). The genetic component of the corolla surface was positively correlated with pollinator visitation rates in urban populations, but not in natural populations (Pollinator Visitation Rate $\times$ Habitat: $F_{(1,11)} = 14.999$, $P = 0.002$; [Figure S7](#), [Table S10](#)).

Urbanization affects the sepal shape of *I. capensis* through plastic responses

The spur of urban flowers was shorter and less curved compared to the ones in natural habitats in the field ($F_{(1,1518)} = 3.751$, $P = 0.019$; [Figure 5B](#), [5C](#), [Table S11A](#)), but not in controlled conditions ($F_{(1,315)} = 0.858$, $P = 0.502$; [Figure 5B](#), [5C](#), [Table S11A](#)). In contrast, corolla shape was similar in urban and natural habitats, both in the common garden ($F_{(1,314)} = 0.436$, $P = 0.837$; [Figure 5E](#), [Table S11A](#)) and in the field ($F_{(1,1509)} = 0.886$, $P = 0.472$; [Figure 5F](#), [Table S11A](#)). Corolla shape of flowers from the field populations also differed from those raised in the common garden (Environment: $F_{(1,1839)} = 6.198$, $P = 0.002$; [Figure 5E](#), [5F](#),

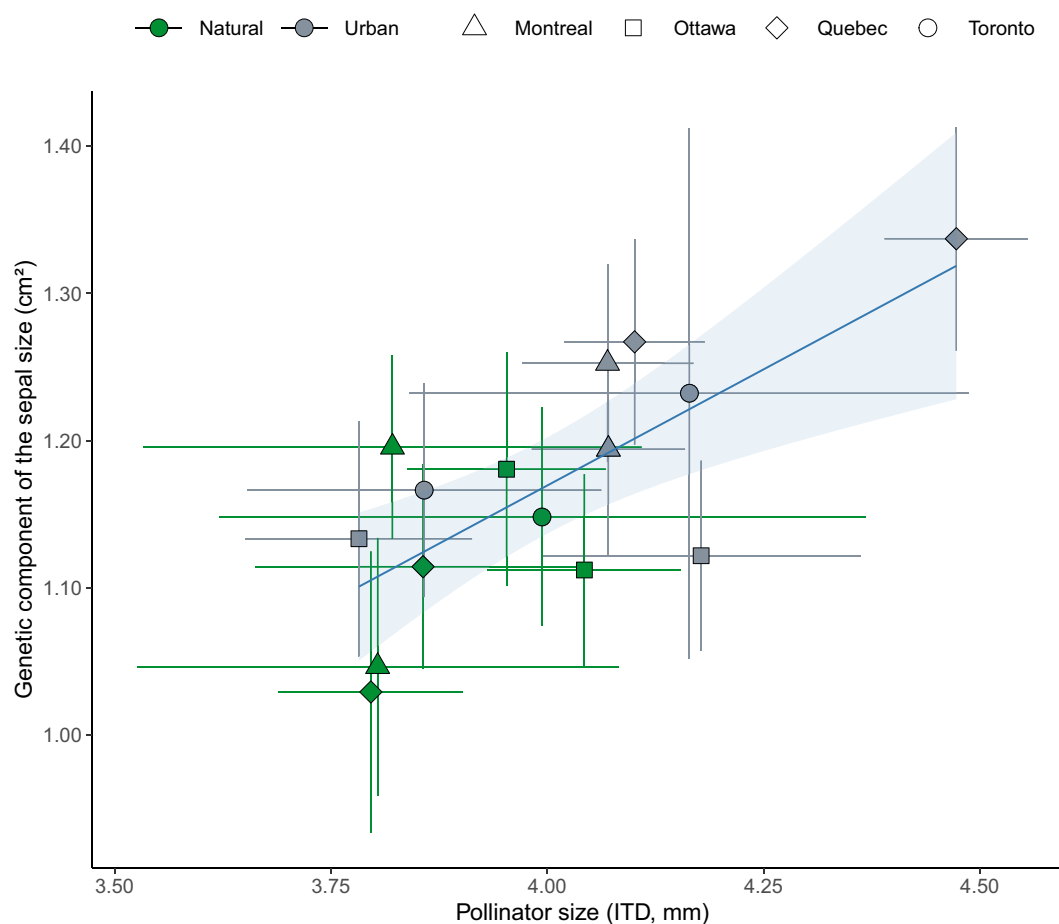


Figure 4. Correlation between the genetic component of the sepal size and bumblebee inter-tegular distances. The points represent the population mean of the sepal size in the common garden experiment and the population mean bumblebee size averaged across the 2021 and 2022 field seasons. Error bars for the points indicate the 95% confidence intervals of the means. The line shows the significant positive linear relationship with its standard error. Measures of bumblebee size include all *Bombus* spp. collected.

Table S11B), further supporting the effect of phenotypic plasticity on flowers. The shape of the corolla in controlled conditions suggests that they are less open than those in the field (Figure 5E, 5F).

Discussion

The combination of fieldwork and common garden experiments across four cities in southeastern Canada revealed that urbanization altered the pollinator communities of *I. capensis* and its flower morphology. Different species of pollinators were associated with urban and natural populations of *I. capensis*. Moreover, urbanization affected bumblebee size differently among regions, a result explained by both intraspecific and interspecific differences. These differences in pollinator community and size support our first hypothesis. The genetic component of sepal size, which is involved in the mechanical fit with pollinators, was larger in urban areas, despite variation among regions. In addition, and supporting our second hypothesis, we found that the genetic component of sepal size was positively correlated with bumblebee size across populations, in both urban and natural habitats. Although we didn't assess fitness in this study, this suggests that sepal size

is adapted to the size of its bumblebee pollinators in both urban and natural habitats despite substantial phenotypic plasticity in the field and regional discrepancies. This positive correlation may be due to improved pollen transfer when the size of the pollinator matches the size of the sepal, although further studies are needed to test this hypothesis.

The impact of urbanization on sepal size varied among regions, but it aligned with the impact of urbanization on pollinator communities. For example, pollinators were the largest in urban populations of the Quebec and Montreal regions, for which the genetic component of sepal size was also the largest. In the Ottawa region, urban and natural pollinators were of similar sizes, and the trend was the same for the genetic component of sepal size. In the Toronto region, bumblebees were smaller in the urban populations, but the genetic component of sepal size did not follow this pattern. Interestingly, this trend in urban-natural pollinator and sepal size differences among regions parallels the environmental variation among them (Figure 1B, 1C): the more the region is situated to the northeast, the larger the bumblebees and the sepals tend to be in urban populations compared to natural ones, suggesting the abiotic environment of these regions may influence observed patterns. Furthermore, fragmentation or heat island effects have been shown to cause variation in

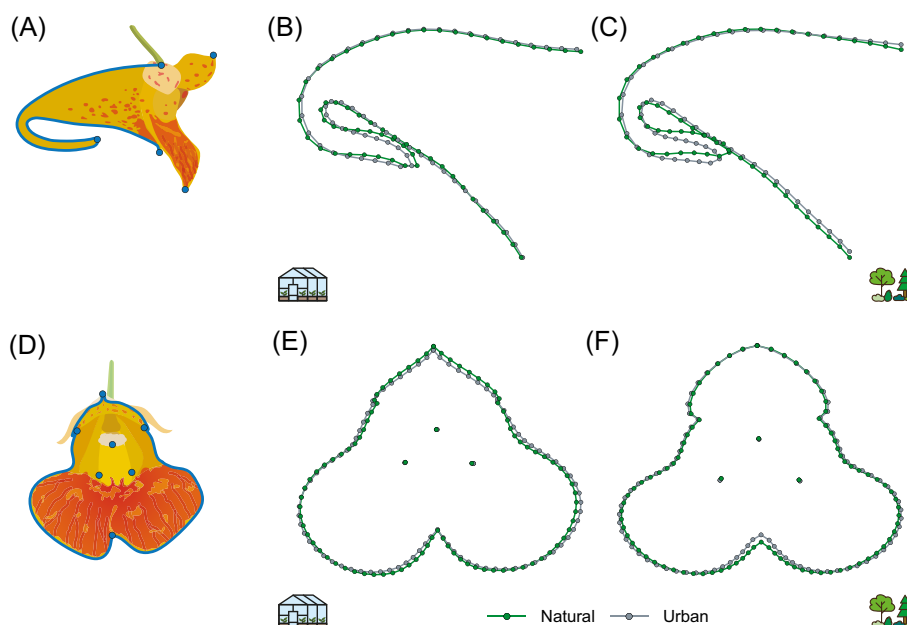


Figure 5. Predicted posterior sepal and corolla shape comparison between habitats. (A) Drawing showing the landmarks (dots) and semilandmarks (curves) position on the posterior sepal of *Impatiens capensis*. (B) Predicted shape of the posterior sepal in urban and natural habitats for the common garden experiment. (C) Predicted shape of the posterior sepal in urban and natural habitats for the 2021 field season. (D) Drawing showing the landmarks (dots) and semilandmarks (curves) position on the corolla of *Impatiens capensis*. (E) Predicted corolla shape in urban and natural habitats for the common garden experiment. (F) Predicted corolla shape in urban and natural habitats for the 2021 field season. Credits for the drawings: J. Lafond.

bee body sizes (Eggenberger et al., 2019; Greenleaf et al., 2007; Theodorou et al., 2021) and the intensity of these urban phenomena among regions may explain the variation of the urbanization effect on bumblebee size across regions. In turn, variation in pollinator size may have influenced floral traits. Previous studies on flower evolution in urban areas have focused on phenotypic changes associated with pollinator availability and pollen movement (Irwin et al., 2018; Ushimaru et al., 2014), but they left aside a more thorough characterization of pollinator communities. Here, we show that characterizing the pollinator community helps to understand the floral evolution in an urbanization context.

Urbanization affected the genetic component of the corolla surface differently among regions, with corollas being larger in Quebec's urban areas. However, corolla surface was positively correlated with pollinator visitation rates in urban populations (Figure S7), contrary to our expectation that larger corollas should be favored in habitats with reduced pollination services (third hypothesis). This finding does not preclude the expected negative relationship between floral display and pollen limitation as our analysis was limited to a single component of floral display, corolla surface, excluding the effect of flower number, and did not account for pollen transfer effectiveness and pollen quality in the estimation of pollen limitation. Trade-offs may also play a role; for example, limited resources could constrain investment in both flower size and number, such that increased flower number compensates for smaller corollas in overall display. Pollinator visitation rates to *I. capensis* may also be driven by the entire plant community through both competitive and facilitated interactions with co-flowering species. For instance, the presence of the large-flowered invasive *Impatiens glandulifera* has been associated with

selection for reduced corolla height in *I. capensis* (Beans & Roach, 2015).

Urbanization had a limited effect on flower shape, partly supporting our fourth hypothesis. In the field, urban sepals had shorter and less curved nectar spurs, which partially supports the results of Faure et al. (2023), who found that urban nectar spurs were less curved in the region of Montreal. However, the lack of difference in the common garden experiment suggests that urbanization affected sepal shape only through phenotypic plasticity, although the statistical power to detect a genetic response was small for the common garden experiment (Table S1). In contrast, corolla shape was not affected by urbanization in both field and common garden conditions. Overall, flower shape—excluding size—shows no evidence of genetic adaptation and seems to be strongly influenced by environmental factors. These findings may seem surprising, as spur curvature, a component we characterize through sepal shape, has been linked to improved pollen removal by hummingbirds (Travers et al., 2003) and longer visit durations by bees (Young, 2008). However, in our populations, bumblebees were the main pollinators, and hummingbirds were rare, potentially limiting their role as selective agents. Finally, there is no reason to believe that visit duration by bumblebees differs between urban and natural habitats, so the lack of difference in our common garden experiment appears logical.

Phenotypic plasticity can be adaptive or maladaptive depending on whether it moves phenotypes towards or away from the fitness optimum of populations. It can also decrease the efficiency of natural selection as the phenotypes with the highest fitness represent both the genetic and plastic components of the corresponding traits. We found that for the sepal size of *I. capensis*, phenotypic plasticity masked the genetic differences detected in the common

garden experiment (Figure 3D; Figure S4), suggesting a counter-gradient variation in which plasticity typically modifies the phenotype in a direction opposite to natural selection (Conover & Schultz, 1995). Assuming a similar plastic response for all genotypes, such as larger flowers in the common garden are also larger in the field conditions, we infer that the sepal size genetically respond to its pollinator community. Therefore, bumblebee selection for sepal size along an urbanization gradient appears to be a case of compensatory adaptive change in which selection counteracts the homogenizing effect of phenotypic plasticity (Conover & Schultz, 1995; Conover et al., 2009; Hendry, 2016). Indeed, natural selection seems to favor larger flowers in some cities because of the presence of larger bumblebees, but plasticity masked the genetic differences in the field (Figure 3D, Figure S4). We also highlighted that the corolla size and shape were significantly affected by the environment through phenotypic plasticity, which is particularly evident from the observed differences between the common garden and field conditions. This plasticity could be caused by developmental constraints, weather conditions, and pollinator visits (by pushing the petals apart), and may at least partly explain the variation among years (Table S6C), and between field and controlled conditions. We acknowledge that because the common-garden experiment involved plants grown from seeds collected in the field, maternal effects could be expressed and confounded with the genetic differentiation we detected. Nevertheless, our results demonstrate that it is critical to disentangle plastic from genetic responses to fully understand the causes of phenotypic changes and adequately interpret the evolutionary responses of organisms to environmental changes (de Villemereuil et al., 2016; Diamond & Martin, 2021; Lambert et al., 2021; Thompson et al., 2025).

In conclusion, we found some general effects of urbanization across the four cities investigated (e.g., pollinator communities, sepal size), although for most traits studied, the effect of urbanization varied among cities. These results are consistent with a previous large-scale experiment that highlighted substantial variation in the response of the white clover to urbanization among cities (Santangelo et al., 2022), suggesting a multifactorial origin to phenotypic changes. Strong environmental changes associated with urbanization impact living organisms, their interactions, and their evolution. Although our knowledge of eco-evolutionary dynamics in these environments has increased in the last decade, integrative approaches that assess plastic and genetic components of traits while considering species interactions are still scarce, but they are necessary to understand rapid evolution to changing environments. Only with this will we be able to understand adaptation to urbanization and pave the way for conservation policies in these habitats.

Supplementary material

Supplementary material is available online at *Evolution Letters*.

Data and code availability

All code and data are available in Dryad at <https://doi.org/10.5061/dryad.931zcrk17>

Author contributions

Conceptualization: S.J., J.B.; Methodology: J.B., S.J., C.D.; Investigation: J.B., J.C., S.C., J.L., E.N., S.J.; Visualization: J.B., S.J., C.D., J.L.; Funding acquisition: S.J.; Project administration: J.B., S.J.; Writing—original draft: J.B.; Writing—review & editing: J.B., J.C., S.C., J.L., E.N., C.D., S.J.

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Conflicts of interest

Authors declare that they have no competing interests.

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